



AKGEC SKILLS FOUNDATION

Training Program Details

FAB LAB CENTRE OF EXCELLENCE

Training/Course Title:	Winter Training Program 3D Printing
Target Audience:	Students – B. Tech / Diploma
Fee:	Rs. 7000
Duration:	40 Hrs.
Module	▪ Safety instructions and FABLAB orientation for 3D printing ▪ Introduction to Additive Manufacturing and 3D printing applications ▪ Overview of 3D printing technologies (FDM – primary focus, SLA – introduction) ▪ 3D printer components and working principles ▪ Types of 3D printing materials and their properties (PLA, ABS, PETG – basics) ▪ Introduction to 3D design and CAD modeling ▪ 3D modeling using CAD software (TinkerCAD / Fusion 360 – basics) ▪ Preparing 3D models for printing (STL export and model checks) ▪ Printer setup, bed leveling, and calibration ▪ Hands-on 3D printing process and job monitoring ▪ Common printing defects and basic troubleshooting ▪ Post-processing techniques (support removal and surface finishing) ▪ Introduction to Design for Additive Manufacturing (DfAM)
Outcome:	▪ Demonstrate safe operation and handling of 3D printing equipment ▪ Understand the fundamentals of additive manufacturing and FDM technology ▪ Design simple 3D models using CAD tool ▪ Prepare, slice, and print 3D models independently ▪ Calibrate and operate a desktop 3D printer ▪ Identify and correct common 3D printing issues ▪ Apply basic DfAM principles for better print quality ▪ Develop a functional prototype using 3D printing
Certification:	Upon successful completion, participants will be awarded a Globally recognized joint certification from FAB LAB and AKGEC.

Program Coordinator:	Name: Mr. Himanshu Tripathi	Contact number: 9873812070
Payment Mode:		